

The Random Furstenberg Set is a $\Lambda(q)$ -Set for Every Finite $q > 2$

Candidate full solution produced in run `fa_banach_001`
Model: GPT5.6

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Status. Candidate full solution, currently assessed as likely valid and awaiting human review. The novelty search described below was bounded, not exhaustive.

1 Source and open question

Fan, Queffélec, and Queffélec define the random Furstenberg set $T = T(\omega)$ by independent selectors

$$\mathbb{P}(n \in T) = \delta_n := \frac{\log n}{n}, \quad n \geq 2. \quad (1)$$

In arXiv:2104.08944v2 they prove that T is almost surely p -Rider for $p > 4/3$ and then ask whether T , like the deterministic Furstenberg set, is a $\Lambda(p)$ -set for every finite p [1, p. 9].

This second result, weaker than its deterministic analogue (Nair), seems new and could possibly be improved.

Theorem 1.7. *The random set T is a.s. a p -Rider set for $p > 4/3$.*

The above first two results are valid for more general random sets that we also consider in Section 5. The third result, not shared by all random sets (see [52]), acts as an incentive for pursuing with S itself. On the other hand, S is a $\Lambda(p)$ -set for all $p < \infty$ (see Section 4, Theorem 4.12), but we do not know the answer as concerns T . However, it can be proved that T contains a relatively big subset which is a $\Lambda(p)$ -set for all $p < \infty$ (We will come back to this question in a forth coming paper).

2. FIRST RESULTS ON THE FURSTENBERG SET

Figure 1: The open question in arXiv:2104.08944v2, PDF page 9, immediately after Theorem 1.7. The crop retains the full readable text width.

For $q > 2$, a set $E \subset \mathbb{Z}$ is a $\Lambda(q)$ -set if there is a constant $C_{q,E} < \infty$ such that every trigonometric polynomial $f(x) = \sum_{n \in E} a_n e^{2\pi i n x}$ satisfies

$$\|f\|_{L^q(\mathbb{T})} \leq C_{q,E} \left(\sum_{n \in E} |a_n|^2 \right)^{1/2}. \quad (2)$$

2 Proof intuition

The selector means in (1) are just large enough to give $|T \cap [1, N]|$ order $(\log N)^2$, but they are small enough that every *fixed* additive-relation type occurs only finitely often. Indeed, if a fixed

integer relation has largest entry M , another entry must be comparable to M . Once the other entries are fixed, this second large entry is determined. The expected number of such relations is then bounded by a convergent series of order $\sum_M (\log M)^{2r-2}/M^2$.

The key bookkeeping point is that a Fourier moment contains repeated indices. Those repetitions cannot be fed directly into the independence calculation. Instead, cancel the common multiplicity of each frequency on the two sides of the moment equation. The remaining distinct support is one of finitely many random relation cores. The canceled part is an arbitrary common multiset, but summing it produces exactly a power of the coefficient ℓ^2 norm. Thus every even Fourier moment has the required bound.

3 Main theorem

Theorem 1. *Let T be the random subset of the positive integers defined by the independent selectors (1). With probability one, simultaneously for every finite $q > 2$, the set T is a $\Lambda(q)$ -set. Equivalently, almost surely, for every finite $q > 2$ there is a random constant $C_{q,T} < \infty$ for which (2) holds for all finitely supported coefficient families on T .*

4 Proof

We first isolate the probabilistic input. A tuple is called injective if all its entries are distinct.

Lemma 1 (Only finitely many fixed injective relations). *Fix $r \geq 1$ and nonzero integers $c_1, \dots, c_r$. Almost surely there are only finitely many injective tuples $(v_1, \dots, v_r) \in T^r$ satisfying*

$$c_1 v_1 + \dots + c_r v_r = 0. \quad (3)$$

Moreover, the almost-sure event can be chosen simultaneously for all such finite integer coefficient vectors.

Proof. There is nothing to prove for a coefficient vector admitting no positive solution, so suppose a solution is given and put $M = \max_k v_k$. Choose an index i with $v_i = M$. The relation gives

$$|c_i| M \leq \sum_{k \neq i} |c_k| v_k.$$

Consequently some $j \neq i$ satisfies $v_j \geq \kappa M$, where $\kappa > 0$ depends only on the fixed coefficient vector. There are only finitely many choices for the ordered pair (i, j) .

Write $D(M) = \sum_{n \leq M} \delta_n$. For M large enough,

$$D(M) \leq C(1 + \log M)^2, \quad \delta_M \leq C \frac{1 + \log M}{M}, \quad \sup_{\kappa M \leq n \leq M} \delta_n \leq C_\kappa \frac{1 + \log M}{M}. \quad (4)$$

Fix i, j , the value $M = v_i$, and all v_k with $k \neq i, j$. Equation (3) determines v_j uniquely. Restricting to injective tuples lets us multiply the selector probabilities. Therefore the expected number N_c of injective solutions is bounded by

$$\begin{aligned} \mathbb{E} N_c &\leq O_c(1) + C_c \sum_{M \geq 3} \delta_M \frac{1 + \log M}{M} D(M)^{r-2} \\ &\leq O_c(1) + C'_c \sum_{M \geq 3} \frac{(1 + \log M)^{2r-2}}{M^2} < \infty. \end{aligned}$$

For $r = 1$ there are no positive solutions, and for $r = 2$ the factor $D(M)^{r-2}$ is interpreted as 1.

Thus $N_c < \infty$ almost surely. There are only countably many finite integer coefficient vectors, so intersecting their probability-one events proves the simultaneous assertion. $\square$

We now work on the probability-one event supplied by Lemma 1. Fix an integer $m \geq 2$ and a trigonometric polynomial

$$f(x) = \sum_{n \in T} a_n e^{2\pi i n x}, \quad B := \sum_{n \in T} |a_n|^2 < \infty.$$

Fourier orthogonality gives

$$\|f\|_{2m}^{2m} = \sum_{x_1 + \dots + x_m = y_1 + \dots + y_m} a_{x_1} \dots a_{x_m} \overline{a_{y_1} \dots a_{y_m}}, \quad (5)$$

where all indices lie in T and in the finite support of f .

For a tuple in (5) and each distinct frequency v , let

$$c_v = \#\{i : x_i = v\} - \#\{j : y_j = v\}.$$

Delete the frequencies with $c_v = 0$. If none remain, call the tuple diagonal. Otherwise the remaining distinct frequencies form an injective reduced core

$$\sum_{v \in V} c_v v = 0, \quad 0 < |c_v| \leq m, \quad \sum_{v \in V} c_v = 0. \quad (6)$$

For fixed m there are only finitely many possible coefficient patterns in (6); Lemma 1 says that each has only finitely many realizations in T . Hence the total number $K_m(T)$ of nonempty reduced cores is finite.

Fix one core and put

$$s = \sum_{c_v > 0} c_v = - \sum_{c_v < 0} c_v \leq m.$$

Every full tuple reducing to this core is obtained by adding a common multiset of $m - s$ frequencies to its two sides and then ordering the slots. To obtain an upper bound, represent that multiset by an ordered list $z_1, \dots, z_{m-s} \in T$ and allow all slot permutations. The total absolute contribution of the fixed core to (5) is at most

$$\begin{aligned} (m!)^2 \left(\prod_{v \in V} |a_v|^{c_v} \right) \sum_{z_1, \dots, z_{m-s} \in T} \prod_{\ell=1}^{m-s} |a_{z_\ell}|^2 &= (m!)^2 \left(\prod_{v \in V} |a_v|^{c_v} \right) B^{m-s} \\ &\leq (m!)^2 B^m. \end{aligned}$$

The last inequality uses $|a_v| \leq B^{1/2}$ and $\sum_v |c_v| = 2s$. The diagonal tuples obey the same estimate with the empty core. Taking absolute values termwise in (5) therefore gives

$$\|f\|_{2m}^{2m} \leq (m!)^2 (1 + K_m(T)) B^m. \quad (7)$$

Thus T is a $\Lambda(2m)$ -set for every integer $m \geq 2$.

Finally, given any finite $q > 2$, choose m with $2m \geq q$. Haar measure on $\mathbb{T}$ is normalized, so $\|f\|_q \leq \|f\|_{2m}$. Combining this with (7) proves the $\Lambda(q)$ inequality and completes the proof of Theorem 1.

5 Verification notes

Two tempting shortcuts would be false, and the proof avoids both. First, a full $2m$ -term relation need not occur only finitely often: a fixed nontrivial core can be padded by arbitrary common frequencies. The proof sums this padding as B^{m-s} . Second, repeated indices cannot be assigned a product of selector probabilities. Independence is invoked only after the support has been reduced to distinct core frequencies.

The relation-count estimate is uniform enough for simultaneous use because there are countably many coefficient vectors. For any fixed moment order only finitely many of those vectors are relevant. No computational experiment is an ingredient of the proof.

6 Novelty search, scope, and limitations

The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2104.08944, the exact title, random Furstenberg sets, $\Lambda(q)$, p -Rider, quasi-independent relations, and selector means $\log n/n$. No duplicate was found.

The authors’ 2023 follow-up still states that a positive-density subset $T' \subset T$ is $\Lambda(q)$ for every finite q , but that they do not know whether T itself has this property [2, Section 5.2]. Bounded arXiv-focused and general web searches on 19 July 2026 used the exact titles, authors, quoted question, and close selector formulations. They found the 2021 source and 2023 follow-up but no later resolution. This is not an exhaustive bibliographic guarantee.

The theorem proves the qualitative $\Lambda(q)$ property with an almost-sure constant depending on q and the realization. It does not provide a useful deterministic bound on that random constant, and it does not settle whether T is 4/3-Rider or whether the deterministic Furstenberg set S is 4/3-Rider or 4/3-Sidon.

7 Human-review recommendation

This is a candidate full affirmative answer to the source’s whole-set $\Lambda(q)$ question. The highest-value expert checks are (i) the summation over a coefficient-dependent comparable pair in Lemma 1; and (ii) the claim that every moment tuple is overcounted by one reduced core, an ordered common list, and at most $(m!)^2$ slot permutations.

References

- [1] A. Fan, H. Queffélec, and M. Queffélec, *The Furstenberg set and its random version*, arXiv:2104.08944v2, 2022.
- [2] A. Fan, H. Queffélec, and M. Queffélec, *Old and new results on the Furstenberg sets*, arXiv:2303.06850v1, 2023.